1. Sketch the region of integration and evaluate

$$\int_0^2 \int_0^{x/2} 6xy^2 \, dy \, dx.$$

2. Sketch the region of integration and evaluate

$$\int_0^1 \int_y^1 x\cos(x^3) \, dx \, dy.$$

3. A solid lies inside the cylinder $x^2 + y^2 = 4$, above $z = 0$, and below $z = 3 + x$. Sketch the solid or a labeled cross-section. Set up, but do not evaluate, a triple integral for its volume in a coordinate system of your choice. Then write an equivalent setup in a second coordinate system.

4. A solid lies inside the sphere $x^2 + y^2 + z^2 = 4$ and above $z = 1$. Sketch the solid or a labeled cross-section. Set up, but do not evaluate, a triple integral for its volume in a coordinate system of your choice. Then write an equivalent setup in a second coordinate system.

5. A solid lies above $z = 0$ and below $z = 4 - x^2 - y^2$, with density $\delta(x, y, z) = 1 + z$. Sketch the solid or a labeled cross-section. Set up, but do not evaluate, a triple integral for its mass in a coordinate system of your choice. Then write an equivalent setup in a second coordinate system.

6. A solid lies inside the sphere $x^2 + y^2 + z^2 = 4$ and above the cone $z = \sqrt{x^2 + y^2}$, with density $\delta(x, y, z) = x^2 + y^2 + z^2$. Sketch the solid or a labeled cross-section. Set up, but do not evaluate, a triple integral for its mass in a coordinate system of your choice. Then write an equivalent setup in a second coordinate system.

7. A wire follows

$$\vec{r}(t) = (2\cos t, 2\sin t, t^2), \qquad 0 \le t \le \frac{\pi}{2},$$

and has density $\delta(x, y, z) = 1 + z$. Set up, but do not evaluate, a line integral representing its mass.

8. A wire follows

$$\vec{r}(t) = (t, t^2, e^{2t}), \qquad 0 \le t \le 1,$$

and has density $\delta(x, y, z) = x^2 + z$. Set up, but do not evaluate, a line integral representing its mass.

9. Find the work done by $\vec{F}(x, y) = (y, x^2)$ along the parabola $y = x^2$, from $(0, 0)$ to $(1, 1)$. Parametrize the curve, set up the line integral, and evaluate it.

10. Find the work done by $\vec{F}(x, y) = (-y, 2x)$ around the ellipse $x^2/4 + y^2 = 1$, traversed once **counterclockwise**. Parametrize the curve, set up the line integral, and evaluate it.

11. Let C be the curve given below, oriented by increasing t:

$$\vec{F}(x, y) = \left(2xy + ye^{xy},\ x^2 + xe^{xy}\right),$$
$$\vec{r}(t) = \left(t + \sin(2\pi t),\ t^2 + \sin(\pi t)\right), \qquad 0 \le t \le 1.$$

(a) Determine whether $\vec{F}$ is conservative and justify your answer.

(b) Using part (a), evaluate $\displaystyle\int_C \vec{F} \cdot d\vec{r}$.

12. Let C be the curve given below, oriented by increasing t:

$$\vec{F}(x, y) = \left(x^2 + y^2,\ 2xy + \cos y\right),$$
$$\vec{r}(t) = \left(1 + \sin(2\pi t),\ \frac{\pi t}{2} + t(1-t)e^t\right), \qquad 0 \le t \le 1.$$

(a) Determine whether $\vec{F}$ is conservative and justify your answer.

(b) Using part (a), evaluate $\displaystyle\int_C \vec{F} \cdot d\vec{r}$.

13. Which of the following integrals evaluate to a finite real number, with no variables remaining? **Select all that apply.** You do not need to compute the numerical values.

(a) $\displaystyle\int_0^1 \int_0^z \int_0^{y+z} (1+x)\ dx\ dy\ dz$

(b) $\displaystyle\int_0^1 \int_0^z \int_0^x 1\ dx\ dy\ dz$

(c) $\displaystyle\int_0^{2\pi} \int_0^1 \int_0^{1-r^2} r\ dz\ dr\ d\theta$

(d) $\displaystyle\int_0^{2\pi} \int_0^z \int_0^1 r\ dz\ dr\ d\theta$

(e) $\displaystyle\int_0^{2\pi} \int_0^{\phi} \int_0^1 \rho^2 \sin\phi\ d\rho\ d\phi\ d\theta$

(f) $\displaystyle\int_0^{2\pi} \int_0^{\pi/2} \int_0^{2\cos\phi} \rho^2 \sin\phi\ d\rho\ d\phi\ d\theta$

(g) $\displaystyle\int_0^1 \int_0^x \int_0^1 (1+y)\ dx\ dy\ dz$

(h) $\displaystyle\int_0^1 \int_0^{1-x} \int_0^{x+y} (1+z)\ dz\ dy\ dx$

(i) $\displaystyle\int_0^{2\pi} \int_0^1 \int_0^{1-r^2} r(1+r^2)\ dz\ dr\ d\theta$

(j) $\displaystyle\int_0^{2\pi} \int_0^1 \int_0^{1-r^2} 1\ dz\ dr\ d\theta$

(k) $\displaystyle\int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^2 \sin\phi\ d\rho\ d\phi\ d\theta$

(l) $\displaystyle\int_0^{2\pi} \int_0^{\rho} \int_0^1 \rho^2 \sin\phi\ d\rho\ d\phi\ d\theta$

14. For each pair $\vec{F}$ and C below, mark the following statement **True or False** and give a brief reason: *The Fundamental Theorem of Line Integrals can be used to evaluate* $\int_C \vec{F} \cdot d\vec{r}$ *from a potential for the given field and the endpoints of* C. Do not evaluate the integrals.

(a) $\vec{F}(x, y) = (2xy, x^2 + 2y)$; C is the line segment from $(0, 0)$ to $(1, 2)$.

(b) $\vec{F}(x, y) = (-y, x)$; C is the unit circle $x^2 + y^2 = 1$, traversed once counterclockwise.

(c) $\vec{F}(x, y) = (y, 2x)$; C is the line segment from $(0, 0)$ to $(1, 1)$.

(d) $\vec{F}(x, y) = (ye^x, e^x)$; C is the unit circle $x^2 + y^2 = 1$, traversed once counterclockwise.